

METHODS OF USE AND OPERATIONAL EMBODIMENTS

In various embodiments, the disclosed intraluminal compliance and distance profiling device is configured for session-based operation in which a user performs a guided measurement procedure to acquire repeatable physiologic data. Operational workflows may vary depending on device configuration, sensing modality, and computational integration level.

In a representative method of use, a user inserts the insertable measurement section into a body lumen until a positional reference element such as a flange or stop contacts an external anatomical reference location. The reference element establishes a repeatable insertion depth to improve measurement consistency between sessions.

Following insertion, the distal sensing region acquires one or more distance measurements directed toward an internal anatomical boundary. Distance acquisition may occur automatically upon stabilization, upon user command, or according to a guided software sequence. Measurements may be averaged across multiple samples to improve robustness against motion or environmental variability.

In embodiments including a compliance characterization region, tissue interaction measurements may be obtained through passive pressure sensing, controlled expansion of an inflatable structure, or hybrid sensing approaches. Inflation-based measurements may involve manual or automated expansion to one or more target conditions while recording pressure response, expansion dynamics, or stabilization behavior.

Operational software may guide the user through measurement steps using visual, auditory, or haptic feedback. Guidance routines may ensure correct orientation, sufficient stabilization time, acceptable motion limits, or completion of required measurement stages prior to data recording.

During operation, onboard electronics or associated computing systems may perform real-time signal validation, motion filtering, and artifact rejection. Measurements determined to be unstable or inconsistent may be discarded or repeated automatically to improve data quality.

In certain embodiments, measurement sessions may generate structured datasets representing distance metrics, compliance curves, deformation response characteristics, stability indicators, or other derived parameters. These datasets may be associated with a persistent user profile to enable longitudinal comparison across repeated sessions.

Methods of operation may further include synchronization with external sensing nodes, calibration routines, or coordinated measurement sequences designed to correlate intraluminal data with external anatomical or physiologic measurements. Such coordinated workflows enable generation of composite indices derived from multiple sensing modalities.

The device may operate in offline modes storing encrypted session data locally or in connected modes transmitting data securely to an associated application or remote computational platform. User-controlled permissions may govern data retention, sharing, analysis, and visualization.

The operational methods described herein therefore support repeatable, privacy-preserving acquisition of intraluminal physiologic information suitable for consumer wellness applications, research environments, calibration workflows, and integrated multi-node physiologic measurement systems.